Economic Modelling 159 (2026) 107572

Contents lists available at ScienceDirect

Economic Modelling

journal homepage: www.journals.elsevier.com/economic-modelling

# How does Marijuana legislation affect crime? Medical and recreational laws across 50 states☆

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ARTICLE INFO

Dataset link: How Does Marijuana Legislation Affect Crime? Medical and Recreational Laws Across 50 States (Reference data)

*JEL classification:*
K14
K23
K42
I18

*Keywords:*
Marijuana legislation
Crime
Policy spillover
Implementation lag
Synthetic difference-in-differences

ABSTRACT

Novel policies may generate unintended spillovers, particularly when legalizing one activity alters incentives for other forms of crime. Marijuana legalization provides a useful setting to examine such effects, given the staggered adoption of medical and recreational laws across all 50 U.S. states. Using both difference-in-differences and synthetic difference-in-differences approaches, we assess how specification choices shape estimated impacts on property crime. Initial results from both methods suggest that recreational legalization increases property crime; however, once state-specific time trends are incorporated into our preferred synthetic specification, the effect becomes negative and statistically insignificant. Overall, the findings indicate that estimated crime effects are highly sensitive to identification assumptions and do not provide robust evidence of an increase in property crime following legalization, underscoring the importance of careful empirical design in policy evaluation.

## 1. Introduction and overview

> **UN Reclassifies Cannabis as a Less Dangerous Drug.**
>
> A United Nations commission voted on Wednesday to remove cannabis for medicinal purposes from a category of the world's most dangerous drugs, a highly anticipated and long-delayed decision that could clear the way for an expansion of marijuana research and medical use.
>
> *The New York Times*, December 2, 2020.

What are the socioeconomic consequences of marijuana legalization? Some argue that legalization removes a social stigma and spurs economic activity, while others suggest that crime increases. We attempt to tease out the potential unintended consequences of legalization with regard to crime. The time is ripe for our research. Over the past decade, a growing assortment of states in the USA has experienced a staggered adoption of marijuana legalization. To take an illustrative example, on election day 2020 five states voted to legalize either medical or recreational marijuana sales. These included Arizona, Mississippi, Montana, New Jersey, and South Dakota. The five states joined a number of others that had recently loosened restrictions on the sale of marijuana. In 2016, four states—California, Nevada, Maine, and Massachusetts—joined four others (Colorado, Washington, Oregon, Alaska) that had already legalized recreational marijuana. Arkansas, Florida, North Dakota, Ohio, and Pennsylvania also passed medical marijuana laws in 2016. By the end of 2022, thirty-nine states and the District of Columbia had legalized one or both forms of marijuana use. Public sentiment has also changed over the last five decades. Negative attitudes towards marijuana have dropped significantly, while overall acceptance has grown: according to one survey, over two-thirds of people supported legalization by 2023.[1]

☆ We are grateful for comments and feedback from one referee and an editor at Economic Modelling. We benefited greatly from comments from Lingui Han and Rosalie Pacula, from participants at the Conference on Unawareness and Unintended Consequences, and at the Eastern Economic Association annual meetings.

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[1] See https://www.pewresearch.org/short-reads/2024/04/10/facts-about-marijuana/.

https://doi.org/10.1016/j.econmod.2026.107572
Received 10 October 2024; Received in revised form 28 January 2026; Accepted 26 February 2026
Available online 4 March 2026
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It is, of course, valuable to study the performance of a newly formal market, such as the one that has recently emerged for marijuana. It is therefore helpful to fix ideas about the properties of this market, and its relation to other markets. Industry sales are above the median relative to similar industries, and both the number of markets and retail sales have increased steadily over time (see https://mjbizdaily.com/factbook-2020/). It is informative to analyze trading volume in marijuana stocks over the course of 2020, which was an important year leading up to the UN declaration. Interestingly, trading spiked tremendously the week of November 5, likely in reaction to the positive news from the UN and US states discussed above.[2]

Given that we study the impact of legalization on crime, it is useful to ask, What are the potential mechanisms that link marijuana legalization and crime? Several answers come to mind. The first mechanism occurs via changes in aggressive behavior. In the short term if marijuana use increases, this reduces aggression, which in turn may reduce crime (Miczek et al., 1994). At the same time, long term use of marijuana has been shown to increase aggression (Boles and Miotto, 2003; Hoaken and Stewart, 2003). Second, the creation of dispensaries may have diverse social effects. On the one hand dispensaries are cash operations, making them a target for crime; and they create crowds, which could lead to more crime. On the other hand, dispensaries reduce the need for the black market and its associated violence; and crowds can create a natural source of policing (Kepple and Freisthler, 2012). Heightened security at such cash-only dispensaries may also deter crime (Chang and Jacobson, 2017). Third, legalization of marijuana can reallocate police resources to more serious crimes, thereby increasing the reported numbers for such crimes (Makin et al., 2019; Adda et al., 2014).

More formally, the research literature identifies both beneficial and nonbeneficial channels through which marijuana affects crime. The beneficial channel is associated with Gary Becker, and emphasizes that legalization makes other criminal activities unprofitable (Becker, 1968; Becker and Murphy, 2013; Becker, 2014). The Becker hypothesis says that marijuana legalization reduces incentives for crime in general. Before legalization, marijuana sales occur in a black market, along with other criminal activities such as violent and property crimes. After legalization, selling marijuana illegally is no longer profitable. The decreased profitability reduces incentives for individuals to engage in criminal activities related to the illegal drug trade. The nonbeneficial channel posits that legalization tends to increase crime through three possible channels (Goldstein, 1985) First is an economic channel, where addicted individuals use illicit activity to fund their marijuana habit. Second is a systemic channel where, individuals remain loyal to illicit suppliers; and third is a psychopharmacological channel, where individuals suffer cognitive impairment which in turn leads to criminal activity. Given the complexity of legalization, our analysis is similar to previous studies (Pacula and Smart, 2017; Lee, 2025), in that we do not purport to narrow down the precise mechanism at play; we focus instead on estimating the effect and then indicating plausible mechanisms.

More importantly for our research, marijuana legalization presents an interesting case to examine potentially unintended policy consequences, since it is relatively novel, and has also been adopted at different times by different states. From this perspective, the recent flurry of marijuana legalization presents an informative context to examine the anticipated and unanticipated effects of policy. When discussing our results, therefore, we highlight particular effects on crime rates that appear unintended. Our main innovation is that we analyze a broad swathe of states that experienced a staggered legalization of marijuana, using a variety of empirical techniques. We discover that the aftermath of policies features a mix of effects on crime rates, some positive, some negative. We use panel data on 50 states for twenty-five years to estimate the effects of marijuana laws on both violent crime and property crime. We first proceed with a difference in differences approach, which informs us that legalization reduced some, but not all, crimes. This result is largely corroborated by a dynamic estimation, which shows that legalization's effects may take two or more years to accrue. In our main, synthetic difference in differences, approach, we find that medical legalization reduces property crime, while recreational legalization reduces violent crime. We also document the interesting phenomenon that effects on crime, when they do occur, manifest themselves powerfully after several years. This latter finding cautions policy- and lawmakers in the marijuana realm to wait before pronouncing on the effectiveness of legalization. In addition, in line with previous work by Pacula and Smart (2017) and Lee (2025), they should be aware of heterogeneity across multiple dimensions in the relationship between state legalization and crime.

By using empirical approaches of progressive generality, we uncover interesting aspects of legalization in a systematic way, and highlight the possibility of potential sources of bias. In particular, the difference in difference analysis points to a positive effect of recreational legalization on property crime. When we move to our initial synthetic difference in difference analysis (with only state and year fixed effects, in column 3 of Table 5), the impact remains positive. By contrast, in our most comprehensive synthetic difference in difference model (augmented to allow for state specific time trends, in column 4 of Table 5), the effect becomes negative and insignificant. Such patterns highlight the potential sources of bias when using only one empirical approach to analyze the impact of marijuana legalization.

The remainder of our paper is organized as follows. Section 2 reviews extant literature, Section 3 describes our data, Section 4 contains our results and discussion, and Section 5 concludes.

## 2. Related literature

Of course, we are not the first to study this topic, and we are interested in the effects of marijuana legalization for multiple reasons. First, we should think about the overall determinants of crime – we hope to contribute to the vast literature which examines this from many angles. Lovett et al. (2022) is a recent paper that assesses the relationship between policing and crime using data from California, finding, like several papers before it, little evidence of a relationship. From a macroeconomic perspective, Jawadi et al. (2021), among others, find a correlation between unemployment and crime rates in European countries, while (Gutiérrez-Romero, 2024) find that violence is a primary cause of migration out of Guatemala, into Mexico.

Beyond the above research, we build primarily on two important research literatures. First is the extensive literature examining the relationship between substance abuse and crime. Among early studies of the link between general substance abuse and crime, both (Joksch and Jones, 1993) and Carpenter and Dobkin (2015) assess the effect of minimum legal drinking ages on crime. Markowitz and Grossman (1998) highlight the link between alcohol regulation and domestic violence towards children. White and Gorman (2000) provides a comprehensive approach to the relationship between drugs and crime: from historical, methodological, and empirical perspectives. Markowitz (2005) studies the relation between alcohol, drugs, and violent crime. A related study by Yoruk and Lee (2018) examines implications of Sunday law restrictions for social outcomes. These papers provide a general background for our study.

Second is the literature on marijuana legislation and its effects on crime. This research, like ours, takes advantage of legalization to ask novel questions concerning causality. Pacula and Kilmer (2003) examine data on arrests to uncover a positive link between marijuana use and crime. Papers looking at medical marijuana legalization include (Gavrilova et al., 2017), who find that it leads to a decrease in violent crime for states bordering Mexico. In addition, Chu

[2] See https://mjbizdaily.com/post-election-marijuana-stock-price-moves-show-benefit-of-public-liquidity/.

and Townsend (2019) find no substantial effects on either violent or property crime at the national level. Dragone et al. (2019) examine legalization of recreational marijuana, comparing Oregon to Washington, using difference-in-differences and spatial regression discontinuity designs. They find that the policy caused a reduction in certain types of crime.[3] (Hansen et al., 2020) find that legalization of recreational marijuana in Colorado and Washington did not lead to significant increases in traffic fatalities. (Sabia et al., 2024) fna and analyze whether its legalization induces crime and increased use of hard drugs. They discover that legalizing recreational marijuana increases use of marijuana among adults, and reduces drug-related arrests for around three to four years after legalization. They also document that the increased marijuana consumption influences neither violent crime nor harder drug use. Brown et al. (2023) study the benefits and costs of legalizing recreational marijuana consumption. The benefits include job creation and establishment of new industry, while the costs include increased heavy use of marijuana and crime. The authors document that after legalization, state incomes grew by three percent on average, but substance disorders and arrests increased by seventeen and thirteen percent. Much of this important literature focuses on either medical or recreational marijuana legalization, while we add value by including both medical and recreational legalization in all of our analyses.

It is also important to highlight the heterogeneity in the legislative impact of marijuana laws on crime rates. We are particularly inspired by two insightful articles that review key aspects of marijuana legislation and its implications. (Pacula and Chaloupka, 2002) overview the considerable heterogeneity in state-level approaches to medical marijuana issues, which reflect workarounds to the prohibitive federal approaches. The authors identify distinct dimensions of medical marijuana policy, in order to classify states in terms of their willingness to explore medical marijuana use. They suggest that given the substantial heterogeneity across states, future efficacy of marijuana laws will be determined on a case by case basis. More recently, Pacula and Smart (2017) comprehensively study the evolution of state-level marijuana liberalization policies. The authors maintain that scientific evidence on the impact of these policies is inconclusive. Their review successfully addresses related issues, and underscores key limitations of studies that evaluate the effects of decriminalization and legalization on marijuana use. The authors pay special attention to the role of heterogeneity of policies, the timing of the evaluations, and the measures of marijuana use. The study highlights the potential for approaches that can credibly yield insight into social outcomes that arise in response to recent developments in marijuana legalization.

We also build on previous work that has utilized state-of-the art empirical techniques. For example, we allow for implementation lags, and in our main analysis we estimate a synthetic difference in differences model (Arkhangelsky et al., 2021). Chu and Townsend (2019) apply this method to investigate the effects of medical marijuana laws on crime at the national level, and find no substantial effects. The approach is motivated by the potential for conflicting estimates of the cause of crime rates, due to endogeneity concerns. Indeed, some important covariates might empirically show an opposite sign from what theory suggests, due to the deterrence effect in panel data. For example, Lin (2009) documents that size of the police force has a negative impact on crime rates, but then finds a positive relation using two stage least squares estimation. In the synthetic difference-in-differences framework, the main statistic is the average effect on the treatment. The benchmark is established by states that have not legalized marijuana. States that legalized marijuana are considered treatments. If the treatment states' crime rates differ significantly from the benchmark, we conclude there may exist a causal effect of marijuana legalization on crime. We also expand the question of legalization's impact to more recent data. Our results complement those in a recent paper by Lee (2025), who examines the effects of recreational marijuana laws using extended DiD methods and finds heterogeneous effects on property and violent crimes.

## 3. Data

Since we analyze state laws and crime, and need to control for confounding variables, we had to compile our data set from a variety of sources. These data were sometimes reported at different frequencies, and made available at different periods. We therefore ultimately use data for a sample period where we could collect all of the variables and controls. The main data on crime rates come from FBI Uniformed Crime Reports (UCR) arrest data from 1995 to 2019, while the data on recreational and medical use laws come from various state level sources. Our analysis involves both violent crime and property crime. Violent crime is a summary of the following four individual crimes: aggravated assault, murder, rape, and robbery. Similarly, property crime combines the following crimes: larceny, burglary, and motor theft. In addition to these data, we utilize a host of state-level controls including: a dummy variable for political party of the governor; a dummy variable for political party in control of the state senate; poverty rate; unemployment rate; population density; average number of neighboring states that have legalized recreational and medical marijuana; cigarette tax rate; liquor-related arrests; number of police per capita; and real income per capita. We select the data for the controls based on discussions in previous research such as (Pacula and Kilmer, 2003) and Dragone et al. (2019). These data are from the BLS, BEA Census Bureau, and other sources displayed in Table A.1 of the Appendix. We also include state and year fixed effects and use robust clustered standard errors.

Table 1 presents summary statistics for crime rates, expressed as the number of crimes per 100,000 residents. Property crime is much more common than violent crime. The median rate of property crime is 3031.25, more than 8 times larger than the median for violent crimes. When we examine the crime subcategories, murder is least common with a median of 4.57, while larceny is the most common with a median of 2151.68. Within each subcategory, the standard deviations range from around one-third to over one-half the magnitude of the means. This indicates a fair amount of variation in our sample, and supports proceeding with further statistical analysis. At the same time, we should be aware of the large heterogeneity across the subcategories. For example, for violent crime with a median rate of 371.80, the dominant subcategory is aggravated assault, which has a median above 230. The next largest is robbery, which at 92.88 is less than half the size of aggravated assault. The other categories have even lower rates. Consequently, when examining results for property crimes, the aggravated assault observations will dominate other subcategories. Similarly, for property crime the highest subcategory is larceny with a median of 2151.68—more than three times larger than the second highest, which is burglary at 613.35. The heterogeneity across subcategories will have implications for our empirical results.

## 4. Empirical results

We pursue our empirical estimation with several approaches. We start with a difference in differences approach, and then estimate a dynamic model that accounts for implementation lags. Our most important results are from synthetic difference in differences estimation. We discuss each in turn.

[3] Several other studies find a positive relationship between marijuana use and crime and arrest records, using different types of crime data, including Pacula and Kilmer (2003), White and Gorman (2000); and Markowitz (2005). Baker (1998) investigates this question in the context of juveniles in Australia. Wu et al. (2021) focuses on Oregon and finds that the legalization of recreational marijuana increased several types of crime.

**Table 1**
Summary statistics of crime rates.

| Variable | N | Mean | Std Dev | Minimum | Median | Maximum |
|---|---|---|---|---|---|---|
| **Violent Crime** | 1250 | 404.23 | 184.66 | 66.88 | 371.80 | 1070.95 |
| Murder | 1250 | 4.81 | 2.60 | 0.47 | 4.57 | 17.51 |
| Rape | 1250 | 36.54 | 14.43 | 11.16 | 34.09 | 161.64 |
| Robbery | 1250 | 100.78 | 62.45 | 6.15 | 92.88 | 423.13 |
| Assault | 1250 | 262.11 | 132.41 | 34.07 | 232.51 | 766.72 |
| **Property Crime** | 1250 | 3178.21 | 998.09 | 1179.79 | 3031.25 | 7500.12 |
| Burglary | 1250 | 658.02 | 265.14 | 126.28 | 613.35 | 1522.36 |
| Larceny | 1250 | 2220.34 | 653.19 | 911.77 | 2151.68 | 5046.93 |
| Vehicle Theft | 1250 | 299.85 | 171.58 | 28.43 | 261.46 | 1157.66 |

The Table presents summary statistics of the crime rates in the broad areas of Violent Crime and Property Crime. The rates are expressed per 1000,000 residents. Violent Crime is further subdivided into murder, rape, robbery and assault. Property crime is further subdivided into burglary, larceny, and vehicle theft. Data are collected from FBI Uniform Crime Reports.

### 4.1. Difference in differences models

Our difference in differences model includes fixed effects, robust and clustered standard errors, and covariates, at the state level. The benchmark regression model is

$$Y_{it} = \beta_0 + \beta_1 X_{it}^j + \beta_2 Z_{it} + \beta_3 S_i + \beta_4 P_t + \mu_{it}, \tag{1}$$

where $Y$ comprises different crime rates (per 100,00), $X$ is a dummy variable for recreational ($j = R$) and/or medical marijuana ($j = M$) legalization (becomes 1 in the first full year after legalization), $Z$ is a list of control variables—disposable per capita personal income, unemployment rate, the number of police per capita, and population, and other variables as described in Section 3 above —and $S$ and $P$ are state and period fixed effects.

We present our estimation results in Table 2, which shows the effect of marijuana legalization on crime rates. The Table provides three models each for violent and property crime, each model differing by which fixed effects and trends are included. Model (1) specifies state and year fixed effects as well as covariates, in estimating the effect of legalizing marijuana on violent crime. Model (2) augments this specification with a linear time trend, and model (3) adds a quadratic trend. (Throughout the paper, all trends are state-specific). Models (4) to (6) provide similar specifications for the effect on property crimes.

We first discuss the impact on violent crime, focusing on the most comprehensive estimation, from model (3), and examining both overall violent crime and its subcategories. In the first two rows, we see no impact of medical legalization on overall crime, but a strong negative effect from recreational legalization. The negative effect of recreational legalization comes from the subcategories of aggravated assault and robbery. We also observe substantial heterogeneity within subcategories in terms of their response to *medical* legalization. While the overall impact of medical legalization on violent crime is insignificant, this masks the fact that three subcategories – murder, rape, and robbery – have significant impacts. Moreover, the effect on murder and robbery is negative, while the effect on rape is positive, and the effect on aggravated assault is insignificant. As mentioned in Section 3, aggravated assault is the largest component of violent crimes, therefore its insignificant effect dominates the overall violent crime impact—leading to the insignificant impact estimated for violent crime.

In general, these results indicate that legalization reduces crime. The sole exception is rape, which increases in the aftermath of legalization. The US Department of Justice expanded the definition of rape in 2012. While earlier definitions specified rape as male to female and did not allow for cases where the victim was incapacitated by drugs or otherwise, the 2012 definition broadened the scope considerably: it allowed for victim and perpetrator of either gender, and explicitly permitted cases where the victim was unable to give consent because of temporary or physical incapacitation.[4] Thus, after the expanded definition of rape, a larger number of assaults of an intimate nature were automatically classifiable as rape. For example, before the expanded definition, a victim who was incapacitated because of marijuana use would not only be unable to seek relief due to incapacitation, but would face difficulty reporting marijuana use, which was itself illegal. Such issues commonly cause under-reporting of rape (Kane et al., 2024). By contrast, once the 2012 definition came into effect, victims who happened to be under the influence of marijuana would face lower barriers to reporting rape. This could partly explain the positive relationship.

Columns (4) through (6) discuss the impact on property crimes. These results are more straightforward than the above, though not consistent in their direction. Medical marijuana legalization has no effect on overall property crime, although there is a small negative effect on motor vehicle theft. Recreational marijuana legalization has an effect on property crime overall, which is driven mainly by larceny and motor vehicle theft. The positive coefficients indicate that recreational legalization tends to increase property crime overall and within the two subcategories. We will discuss possible channels for these effects in the next Section.

### 4.2. Dynamic effects of Marijuana legalization on crime rates

We now turn to estimation results that reflect the important economic phenomenon of implementation lags. When marijuana use is legalized, there is typically a lag between the date when the law comes into effect, and the date when citizens have access to marijuana. The lag occurs for a number of reasons, including the time to build, submit, and approve dispensaries for marijuana in a particular state. Another source of lags is that it takes time for marijuana purchases to pick up, similar to many newly marketed products. For example, California legalized medical marijuana on November 5th, 1996, so we consider it to be legalized starting from 1997. We therefore express California's two-year lagged effect using the notation $\text{D_MED}_{t+2}$, which equals 1 in 1999 and equals 0 in other years. Similarly, we write a five-year and beyond effect as $\text{D_MED}_{t+5}\text{on}$, which equals 1 from 2002 on and equals 0 otherwise.

Table 3 presents the dynamic impact of legalization on crime. Let us first consider violent crimes, focusing on the most comprehensive model in column (3). For overall violent crimes medical legalization has no lag effect, while recreational legalization does have an effect at the first and second lags. These results support our previous non-dynamic results from Table 2, which demonstrate an impact on overall crime from recreational but not medical legalization. Digging deeper into the subcategories, we see one significant lag effect of recreational legalization on robbery, and two significant lag effects for aggravated assault. For medical legalization, the first, second, and fifth lags for murder are all negative and significant and all except the fourth lag are significant for robbery. However, the much larger category of aggravated assault shows an impact only at the fourth lag for medical legalization. This pattern mirrors the results from Table 2, where murder and robbery subcategories exhibited a significant drop, while aggravated assault and violent crime as a whole showed little impact from medical legalization.

[4] Further details on the expanded definition of rape may be found in the Department of Justice Archives at https://www.justice.gov/archives/opa/blog/updated-definition-rape.

**Table 2**
Difference-in-differences analysis of violent and property crime.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| **Violent Crimes:** | | | | **Property Crimes:** | | |
| D_MED | −0.004 | −0.037*** | −0.004 | −0.007 | 0.006 | −0.007 |
| | (−0.22) | (−3.14) | (−0.34) | (−0.72) | (0.53) | (−0.58) |
| D_REC | −0.006 | −0.045 | −0.068** | 0.044 | 0.051* | 0.042* |
| | (−0.19) | (−1.51) | (−2.48) | (1.58) | (1.78) | (1.95) |
| adj. $R^2$ | 0.9183 | 0.9676 | 0.9756 | 0.9343 | 0.9578 | 0.9735 |
| **1. Murder:** | | | | **1. Burglary:** | | |
| D_MED | −0.011 | −0.055** | −0.066** | −0.009 | −0.010 | 0.008 |
| | (−0.51) | (−2.24) | (−2.23) | (−0.66) | (−0.68) | (0.58) |
| D_REC | −0.048 | −0.091** | −0.002 | 0.056 | 0.057 | 0.028 |
| | (−1.34) | (−2.09) | (−0.04) | (1.43) | (1.45) | (0.99) |
| adj. $R^2$ | 0.9017 | 0.9177 | 0.9217 | 0.9390 | 0.9618 | 0.9767 |
| **2. Rape:** | | | | **2. Larceny Theft:** | | |
| D_MED | 0.025 | 0.050*** | 0.039** | 0.001 | 0.004 | −0.009 |
| | (1.44) | (3.08) | (2.14) | (0.06) | (0.41) | (−0.84) |
| D_REC | −0.001 | 0.021 | −0.004 | 0.041* | 0.063** | 0.038* |
| | (−0.02) | (0.51) | (−0.11) | (1.70) | (2.31) | (1.92) |
| adj. $R^2$ | 0.8423 | 0.9027 | 0.9289 | 0.9247 | 0.9529 | 0.9705 |
| **3. Robbery:** | | | | **3. Motor Vehicle Theft:** | | |
| D_MED | 0.000 | −0.034* | −0.041** | −0.103*** | −0.000 | −0.037* |
| | (0.01) | (−1.86) | (−2.10) | (−4.31) | (−0.00) | (−1.77) |
| D_REC | 0.035 | −0.052 | −0.060* | 0.058 | −0.046 | 0.072* |
| | (1.04) | (−1.32) | (−1.85) | (1.08) | (−1.17) | (1.72) |
| adj. $R^2$ | 0.9641 | 0.9788 | 0.9849 | 0.9102 | 0.9590 | 0.9716 |
| **4. Aggravated Assault:** | | | | | | |
| D_MED | −0.002 | −0.049*** | 0.002 | | | |
| | (−0.11) | (−3.41) | (0.16) | | | |
| D_REC | −0.032 | −0.067** | −0.102*** | | | |
| | (−0.91) | (−2.03) | (−2.99) | | | |
| adj. $R^2$ | 0.8961 | 0.9555 | 0.9692 | | | |
| State and Year Fixed Effects | Yes | Yes | Yes | Yes | Yes | Yes |
| Linear Trend | | Yes | Yes | | Yes | Yes |
| Quadratic Trend | | | Yes | | | Yes |

Models (1) to (3) are for Violent Crimes. Models (4) to (6) are for Property Crimes. All models include state and year fixed-effects, as well as state-level covariates. There are 11 covariates: a dummy variable for political party of the governor; a dummy variable for political party in control of the state senate; poverty rate; unemployment rate; population density; average number of neighboring states that have legalized recreational and medical marijuana; cigarette tax rate; liquor-related arrests; number of police per capita; and real income. Models (2) and (5) add state specific time trends by multiplying the State dummy by $t$. Models (3) and (6) add a state-specific trend $t^2$ in addition to $t$. The sample period is from 1995 to 2019 for 50 states. The number of observations is 1198 for all regressions since several states failed to report crime data to the FBI. D_MED (D_REC) equals 1 if medical (recreational) marijuana is legalized for the entire year, and 0 otherwise. $t$-values are in parentheses.

*** p-values less than 0.01.

** p-values less than 0.05.

* p-values less than 0.10.

We now turn to property crimes, emphasizing Table 3's most comprehensive model in column (6). Overall, the results are roughly similar to those for violent crime: medical legalization has no lag effect while recreational legalization does impact property crimes at all lags. Considering the individual subcategories of property crime, medical legalization has only one small lag effect for auto vehicle theft but otherwise no lag effects for any other subcategories. Recreational legalization has consistent effects at all lags for all subcategories of property crimes. Our results therefore corroborate the heterogeneity argument of previous research (Pacula and Smart, 2017), since we find heterogeneous effects across crimes and legalization types (medical versus recreational). 'F-test for equal D_MED' tests for equal coefficients across different D_MED terms. Let $\beta_j$ denote the estimated coefficient on D_$MED_{t+j}$, for $j = 1, \ldots, 5$. Then the corresponding null hypothesis for this F-test is that the betas are equal, $H_0$: $\beta_1 = \beta_2 = \cdots = \beta_5$. Similarly, 'F-test for equal D_REC' tests for equal coefficients on D_REC terms. These tests provide useful information on whether coefficients statistically differ from each other. For example, the coefficients on medical legalization dummies appear visually close to each other for overall property crimes, with most of them between −0.021 and −0.010 in the most comprehensive model of column (6). By contrast, the coefficients for recreational legalization appear distinct, ranging from 0.045 to 0.322. These different visual patterns are reflected in the F-tests: Table 3's 'F-test for equal D_MED' in column (6) shows a large $p$-value of 0.80 and therefore we cannot reject the null hypothesis of equal coefficients, while 'F-test for equal D_REC' shows strong evidence against the null hypothesis.

A principal reason for examining the F-statistics is to clarify whether the dynamic estimates are meaningfully better than the static model of Table 2. When the coefficients on the multi-year dummies do not statistically differ from each other, this indicates that the single dummy in Table 2 is likely sufficient to capture the main effect. By contrast, when the coefficients on multi-year dummies are statistically significant, then the dynamic specification is more informative.

In sum, lag effects do matter for both violent and property crimes. And in both crime categories overall, such effects show up in the impact of recreational legalization. Medical legalization has no effect overall, but does reduce the violent crime subcategories of robbery and murder. Therefore, the violent crime results are similar to those in the non-dynamic model of Table 2, where the discrepancy between overall and subcategory impact reflects the fact that robbery and murder are relatively small subcategories.

**Table 3**
Dynamic impact of marijuana legalization on crime.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Violent Crimes | | | Property Crimes | | |
| D_MED$_t$ | 0.009 | −0.030* | −0.005 | −0.004 | 0.002 | −0.012 |
| | (0.31) | (−1.90) | (−0.36) | (−0.27) | (0.17) | (−0.93) |
| D_MED$_{t+1}$ | 0.009 | −0.035** | −0.001 | 0.001 | 0.011 | −0.013 |
| | (0.28) | (−2.17) | (−0.08) | (0.06) | (0.74) | (−0.86) |
| D_MED$_{t+2}$ | −0.003 | −0.048*** | −0.004 | −0.004 | 0.014 | −0.017 |
| | (−0.11) | (−3.00) | (−0.20) | (−0.24) | (0.90) | (−1.00) |
| D_MED$_{t+3}$ | −0.038 | −0.044** | −0.000 | −0.014 | 0.009 | −0.010 |
| | (−1.46) | (−2.50) | (−0.00) | (−0.72) | (0.47) | (−0.51) |
| D_MED$_{t+4}$ | −0.018 | −0.030 | 0.016 | 0.009 | 0.033 | 0.005 |
| | (−0.69) | (−1.36) | (0.64) | (0.43) | (1.45) | (0.20) |
| D_MED$_{t+5}$on | −0.000 | −0.058** | 0.012 | −0.022 | 0.007 | −0.021 |
| | (−0.02) | (−2.51) | (0.42) | (−1.38) | (0.32) | (−0.80) |
| D_REC$_t$ | −0.011 | −0.044 | −0.046 | −0.018 | 0.018 | 0.045** |
| | (−0.17) | (−1.07) | (−1.37) | (−0.42) | (0.62) | (2.08) |
| D_REC$_{t+1}$ | −0.038 | −0.057 | −0.059** | 0.032 | 0.040 | 0.073** |
| | (−1.05) | (−1.61) | (−2.00) | (0.59) | (0.79) | (2.17) |
| D_REC$_{t+2}$ | −0.050 | −0.076 | −0.084** | 0.089* | 0.100* | 0.116*** |
| | (−1.43) | (−1.33) | (−1.97) | (1.84) | (1.94) | (2.97) |
| D_REC$_{t+3}$ | −0.011 | −0.024 | −0.042 | 0.125*** | 0.169*** | 0.192*** |
| | (−0.23) | (−0.39) | (−0.98) | (4.94) | (4.49) | (4.52) |
| D_REC$_{t+3}$ | 0.067* | −0.060* | −0.017 | 0.098*** | 0.136*** | 0.216*** |
| | (1.88) | (−1.88) | (−0.35) | (4.15) | (2.80) | (4.53) |
| D_REC$_{t+5}$on | 0.089** | 0.008 | 0.059 | 0.156*** | 0.196*** | 0.322*** |
| | (2.08) | (0.18) | (0.94) | (4.39) | (2.95) | (5.09) |
| adj. $R^2$ | 0.9179 | 0.9674 | 0.9754 | 0.9348 | 0.9583 | 0.9741 |
| F-test for equal D_MED | 0.69 | 0.52 | 0.36 | 0.66 | 0.36 | 0.41 |
| ($p$-value) | (0.60) | (0.72) | (0.84) | (0.62) | (0.84) | (0.80) |
| F-test for equal D_REC | 3.76 | 1.45 | 2.65 | 1.41 | 2.39 | 5.29 |
| ($p$-value) | (0.00) | (0.21) | (0.03) | (0.23) | (0.05) | (0.00) |
| F-test for all D | 2.11 | 0.96 | 2.38 | 3.97 | 2.80 | 3.30 |
| ($p$-value) | (0.03) | (0.47) | (0.01) | (0.00) | (0.00) | (0.00) |
| | 1. Murder | | | 1. Burglary | | |
| D_MED$_t$ | −0.002 | −0.048 | −0.068* | 0.008 | −0.000 | −0.000 |
| | (−0.06) | (−1.33) | (−1.73) | (0.32) | (−0.02) | (−0.03) |
| D_MED$_{t+1}$ | −0.024 | −0.069* | −0.090** | 0.012 | −0.003 | −0.003 |
| | (−0.60) | (−1.83) | (−2.18) | (0.39) | (−0.17) | (−0.19) |
| D_MED$_{t+2}$ | −0.010 | −0.062* | −0.085* | 0.002 | −0.006 | −0.002 |
| | (−0.25) | (−1.68) | (−1.87) | (0.06) | (−0.32) | (−0.13) |
| D_MED$_{t+3}$ | 0.006 | −0.060 | −0.071 | −0.029 | −0.027 | −0.004 |
| | (0.14) | (−1.49) | (−1.40) | (−1.24) | (−1.26) | (−0.21) |
| D_MED$_{t+4}$ | 0.010 | −0.050 | −0.069 | −0.013 | −0.012 | 0.006 |
| | (0.22) | (−1.10) | (−1.15) | (−0.52) | (−0.42) | (0.22) |
| D_MED$_{t+5}$on | −0.032 | −0.141*** | −0.143** | −0.041** | −0.063** | −0.023 |
| | (−0.98) | (−2.92) | (−2.03) | (−2.10) | (−2.19) | (−0.72) |
| D_REC$_t$ | 0.005 | −0.033 | 0.043 | −0.035 | −0.017 | 0.018 |
| | (0.10) | (−0.71) | (0.93) | (−0.58) | (−0.42) | (0.63) |
| D_REC$_{t+1}$ | −0.084 | −0.160*** | −0.056 | 0.063 | 0.055 | 0.083* |
| | (−1.60) | (−3.21) | (−1.00) | (0.72) | (0.86) | (1.94) |
| D_REC$_{t+2}$ | −0.148* | −0.251*** | −0.128 | 0.140** | 0.112* | 0.118** |
| | (−1.96) | (−3.37) | (−1.47) | (2.13) | (1.86) | (2.47) |
| D_REC$_{t+3}$ | −0.003 | −0.143* | −0.078 | 0.144*** | 0.167*** | 0.200*** |
| | (−0.06) | (−1.96) | (−0.91) | (2.83) | (2.86) | (3.76) |
| D_REC$_{t+3}$ | 0.065 | −0.087 | 0.001 | 0.128*** | 0.112* | 0.222*** |
| | (1.41) | (−1.29) | (0.01) | (2.71) | (1.83) | (3.41) |
| D_REC$_{t+5}$on | 0.012 | −0.109 | −0.011 | 0.194*** | 0.191** | 0.348*** |
| | (0.20) | (−1.19) | (−0.08) | (5.82) | (2.54) | (4.31) |
| adj. $R^2$ | 0.9013 | 0.9179 | 0.9215 | 0.9397 | 0.9625 | 0.9771 |
| F-test for equal D_MED | 0.32 | 1.21 | 0.62 | 1.10 | 1.61 | 0.34 |
| ($p$-value) | (0.87) | (0.31) | (0.64) | (0.36) | (0.17) | (0.85) |
| F-test for equal D_REC | 2.69 | 1.18 | 0.92 | 0.95 | 1.03 | 4.14 |
| ($p$-value) | (0.03) | (0.32) | (0.45) | (0.44) | (0.39) | (0.00) |
| F-test for all D | 1.29 | 1.65 | 0.78 | 4.55 | 2.60 | 2.53 |
| ($p$-value) | (0.24) | (0.10) | (0.64) | (0.00) | (0.01) | (0.01) |
| | 2. Rape | | | 2. Larceny Theft | | |
| D_MED$_t$ | 0.065** | 0.062*** | 0.042* | 0.000 | 0.001 | −0.013 |
| | (2.21) | (2.61) | (1.69) | (0.00) | (0.10) | (−0.99) |
| D_MED$_{t+1}$ | 0.036 | 0.044* | 0.033 | 0.005 | 0.010 | −0.015 |
| | (1.30) | (1.86) | (1.47) | (0.34) | (0.70) | (−1.05) |
| D_MED$_{t+2}$ | 0.028 | 0.042** | 0.028 | 0.004 | 0.015 | −0.018 |
| | (1.05) | (2.00) | (1.21) | (0.24) | (1.04) | (−1.20) |
| D_MED$_{t+3}$ | 0.000 | 0.051** | 0.039 | −0.001 | 0.008 | −0.016 |
| | (0.00) | (2.00) | (1.48) | (−0.06) | (0.42) | (−0.89) |
| D_MED$_{t+4}$ | 0.002 | 0.053 | 0.028 | 0.020 | 0.029 | −0.002 |
| | (0.05) | (1.63) | (0.90) | (1.07) | (1.42) | (−0.11) |
| D_MED$_{t+5}$on | −0.016 | 0.057** | 0.034 | −0.011 | 0.017 | −0.025 |
| | (−0.65) | (2.01) | (0.99) | (−0.75) | (0.79) | (−1.07) |
| D_REC$_t$ | 0.019 | 0.017 | −0.015 | −0.004 | 0.038 | 0.040* |
| | (0.37) | (0.29) | (−0.45) | (−0.11) | (1.38) | (1.91) |
| D_REC$_{t+1}$ | 0.068 | 0.048 | 0.075** | 0.030 | 0.047 | 0.054* |
| | (1.14) | (0.94) | (2.04) | (0.61) | (0.96) | (1.88) |
| D_REC$_{t+2}$ | 0.006 | 0.001 | −0.002 | 0.090** | 0.119** | 0.098*** |
| | (0.10) | (0.02) | (−0.04) | (1.98) | (2.17) | (2.74) |
| D_REC$_{t+3}$ | −0.093 | −0.011 | 0.019 | 0.091*** | 0.170*** | 0.158*** |
| | (−1.07) | (−0.24) | (0.33) | (3.82) | (5.20) | (4.18) |
| D_REC$_{t+3}$ | −0.021 | 0.039 | 0.091 | 0.065*** | 0.148*** | 0.163*** |
| | (−0.20) | (0.82) | (1.52) | (3.13) | (3.98) | (3.88) |
| D_REC$_{t+5}$on | −0.040 | 0.068 | 0.162** | 0.133*** | 0.215*** | 0.245*** |
| | (−0.48) | (1.31) | (2.04) | (4.58) | (3.84) | (4.38) |

(*continued on next page*)

**Table 3** (*continued*).

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| adj. $R^2$ | 0.8426 | 0.9020 | 0.9287 | 0.9248 | 0.9535 | 0.9709 |
| F-test for equal D_MED | 1.07 | 0.10 | 0.08 | 0.72 | 0.30 | 0.41 |
| (*p*-value) | (0.37) | (0.98) | (0.99) | (0.58) | (0.88) | (0.80) |
| F-test for equal D_REC | 0.74 | 1.44 | 4.12 | 1.75 | 3.05 | 4.50 |
| (*p*-value) | (0.56) | (0.22) | (0.00) | (0.14) | (0.02) | (0.00) |
| F-test for all D | 0.82 | 0.70 | 1.90 | 2.79 | 4.03 | 2.77 |
| (*p*-value) | (0.60) | (0.71) | (0.05) | (0.00) | (0.00) | (0.00) |
| | 3. Robbery | | | 3. Auto Vehicle Theft | | |
| $D_MED_t$ | 0.015 | −0.018 | −0.033 | −0.091*** | −0.024 | −0.049** |
| | (0.42) | (−0.76) | (−1.50) | (−2.77) | (−1.08) | (−2.01) |
| $D_MED_{t+1}$ | 0.011 | −0.030 | −0.052** | −0.079* | 0.008 | −0.033 |
| | (0.31) | (−1.26) | (−1.96) | (−1.92) | (0.34) | (−1.22) |
| $D_MED_{t+2}$ | −0.007 | −0.038 | −0.067** | −0.119*** | −0.011 | −0.056* |
| | (−0.19) | (−1.45) | (−2.45) | (−2.72) | (−0.38) | (−1.71) |
| $D_MED_{t+3}$ | −0.044 | −0.056* | −0.074** | −0.137*** | 0.019 | −0.010 |
| | (−1.52) | (−1.79) | (−2.11) | (−2.93) | (0.51) | (−0.27) |
| $D_MED_{t+4}$ | −0.002 | −0.019 | −0.056 | −0.127** | 0.036 | −0.007 |
| | (−0.07) | (−0.58) | (−1.46) | (−2.37) | (0.84) | (−0.16) |
| $D_MED_{t+5}$on | 0.016 | −0.059* | −0.075* | −0.091** | −0.012 | −0.030 |
| | (0.53) | (−1.73) | (−1.72) | (−2.49) | (−0.33) | (−0.64) |
| $D_REC_t$ | −0.035 | −0.082** | −0.044 | −0.041 | −0.054 | 0.093** |
| | (−0.61) | (−2.12) | (−1.35) | (−0.59) | (−1.16) | (2.09) |
| $D_REC_{t+1}$ | −0.024 | −0.084 | −0.044 | 0.036 | −0.052 | 0.110* |
| | (−0.46) | (−1.18) | (−1.02) | (0.37) | (−0.73) | (1.75) |
| $D_REC_{t+2}$ | 0.031 | −0.018 | −0.017 | 0.058 | −0.053 | 0.135** |
| | (0.60) | (−0.21) | (−0.28) | (0.51) | (−0.71) | (1.99) |
| $D_REC_{t+3}$ | 0.086** | 0.042 | 0.020 | 0.266*** | 0.071 | 0.260*** |
| | (2.34) | (0.52) | (0.33) | (4.67) | (1.04) | (3.02) |
| $D_REC_{t+3}$ | 0.149** | −0.016 | 0.037 | 0.219* | −0.048 | 0.341*** |
| | (2.22) | (−0.21) | (0.50) | (1.76) | (−0.40) | (3.30) |
| $D_REC_{t+5}$on | 0.229*** | 0.108 | 0.201** | 0.126 | −0.130 | 0.451*** |
| | (5.40) | (1.59) | (2.40) | (1.00) | (−1.09) | (3.79) |
| adj. $R^2$ | 0.9642 | 0.9788 | 0.9849 | 0.9104 | 0.9590 | 0.9719 |
| F-test for equal D_MED | 0.98 | 0.54 | 0.27 | 0.41 | 0.47 | 0.64 |
| (*p*-value) | (0.42) | (0.71) | (0.90) | (0.80) | (0.76) | (0.63) |
| F-test for equal D_REC | 5.80 | 1.83 | 4.76 | 2.02 | 1.29 | 2.92 |
| (*p*-value) | (0.00) | (0.12) | (0.00) | (0.09) | (0.27) | (0.02) |
| F-test for all D | 4.95 | 1.34 | 2.47 | 4.72 | 0.82 | 2.05 |
| (*p*-value) | (0.00) | (0.21) | (0.01) | (0.00) | (0.60) | (0.03) |
| | 4. Aggravated Assault | | | | | |
| $D_MED_t$ | 0.001 | −0.047*** | 0.000 | | | |
| | (0.03) | (−2.71) | (0.02) | | | |
| $D_MED_{t+1}$ | 0.008 | −0.048** | 0.014 | | | |
| | (0.24) | (−2.52) | (0.69) | | | |
| $D_MED_{t+2}$ | 0.001 | −0.060*** | 0.021 | | | |
| | (0.04) | (−2.73) | (0.81) | | | |
| $D_MED_{t+3}$ | −0.028 | −0.048** | 0.031 | | | |
| | (−0.91) | (−2.15) | (1.22) | | | |
| $D_MED_{t+4}$ | −0.008 | −0.035 | 0.062** | | | |
| | (−0.26) | (−1.42) | (2.01) | | | |
| $D_MED_{t+5}$on | 0.003 | −0.062** | 0.074* | | | |
| | (0.09) | (−2.08) | (1.91) | | | |
| $D_REC_t$ | −0.036 | −0.063 | −0.067 | | | |
| | (−0.50) | (−1.46) | (−1.54) | | | |
| $D_REC_{t+1}$ | −0.069 | −0.072* | −0.104*** | | | |
| | (−1.34) | (−1.76) | (−2.76) | | | |
| $D_REC_{t+2}$ | −0.067 | −0.097 | −0.125** | | | |
| | (−1.41) | (−1.43) | (−2.58) | | | |
| $D_REC_{t+3}$ | −0.031 | −0.039 | −0.071 | | | |
| | (−0.59) | (−0.48) | (−1.31) | | | |
| $D_REC_{t+3}$ | 0.042 | −0.094** | −0.054 | | | |
| | (0.92) | (−2.00) | (−0.86) | | | |
| $D_REC_{t+5}$on | 0.055 | −0.032 | 0.009 | | | |
| | (1.34) | (−0.69) | (0.11) | | | |
| adj. $R^2$ | 0.8954 | 0.9552 | 0.9691 | | | |
| F-test for equal D_MED | 0.34 | 0.40 | 1.67 | | | |
| (*p*-value) | (0.85) | (0.81) | (0.15) | | | |
| F-test for equal D_REC | 2.38 | 0.70 | 1.21 | | | |
| (*p*-value) | (0.05) | (0.60) | (0.30) | | | |
| F-test for all D | 1.25 | 0.56 | 2.28 | | | |
| (*p*-value) | (0.26) | (0.83) | (0.02) | | | |

The Table summarizes estimation results for the impact of marijuana legalization on crimes. Models (1) to (3) are for Violent Crimes. Models (4) to (6) are for Property Crimes. All models include state and year fixed-effects, and state-level covariates: a dummy variable for governor's political party; a dummy variable for party in control of state senate; poverty rate; unemployment rate; population density; average number of neighbor states that legalized recreational and medical marijuana; cigarette tax rate; liquor-related arrests; police per capita; and real income. t-values are in parentheses for the main estimation. For F-tests the p-values are in parentheses. F-test for equal D_MED tests for equal coefficients across different D_MED terms. F-test for equal D_REC tests for equal coefficients on D_REC terms. F-test for all tests for equal coefficients on both D_MED and D_REC terms. Models (2) and (5) add state specific time trends by multiplying the State dummy by $t$. Models (3) and (6) add a state-specific trend $t^2$ in addition to $t$. The sample period is from 1995 to 2019 for 50 states. "t+5 on" shows the effect on crime rates 5 or more years later. D_MED (D_REC) equals 1 if medical (recreational) marijuana is legalized for the entire year, and 0 otherwise.

*** p-values smaller than 0.01.

** p-values smaller than 0.05.

* p-values smaller than 0.1.

**Table 4**
Evaluating parallel trends assumption.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Violent Crimes | | | Property Crimes | | |
| $D_MED_{t-2}$ | 0.021 | −0.028* | −0.012 | 0.011 | 0.020 | 0.008 |
| | (0.74) | (−1.79) | (−0.77) | (0.76) | (1.52) | (0.64) |
| $D_MED_{t-1}$ | 0.025 | −0.036** | −0.012 | 0.002 | 0.008 | −0.011 |
| | (0.86) | (−2.01) | (−0.66) | (0.12) | (0.52) | (−0.75) |
| $D_REC_{t-2}$ | 0.018 | 0.020 | 0.016 | −0.025 | 0.016 | −0.021 |
| | (0.34) | (0.82) | (0.62) | (−0.93) | (0.85) | (−1.04) |
| $D_REC_{t-1}$ | −0.007 | 0.002 | −0.014 | −0.052 | 0.001 | −0.003 |
| | (−0.12) | (0.07) | (−0.52) | (−1.62) | (0.05) | (−0.13) |
| F-test for D_MED | 0.57 | 2.81 | 0.35 | 0.29 | 1.17 | 0.80 |
| (*p*-value) | (0.56) | (0.06) | (0.70) | (0.75) | (0.31) | (0.45) |
| F-test for D_REC | 0.07 | 0.36 | 0.77 | 1.61 | 0.39 | 0.61 |
| (*p*-value) | (0.93) | (0.70) | (0.46) | (0.20) | (0.68) | (0.54) |
| F-test for all | 0.32 | 1.69 | 0.59 | 0.96 | 0.72 | 0.70 |
| (*p*-value) | (0.87) | (0.15) | (0.67) | (0.43) | (0.58) | (0.59) |
| $D_MED_{t}$ | 0.017 | −0.045*** | −0.014 | −0.004 | 0.010 | −0.016 |
| | (0.54) | (−2.64) | (−0.71) | (−0.21) | (0.67) | (−0.96) |
| $D_MED_{t+1}$ | 0.017 | −0.053*** | −0.012 | 0.002 | 0.020 | −0.017 |
| | (0.52) | (−2.88) | (−0.50) | (0.10) | (1.19) | (−0.93) |
| $D_MED_{t+2}$ | 0.004 | −0.067*** | −0.017 | −0.002 | 0.022 | −0.020 |
| | (0.12) | (−3.76) | (−0.63) | (−0.09) | (1.33) | (−1.00) |
| $D_MED_{t+3}$ | −0.029 | −0.063*** | −0.012 | −0.011 | 0.020 | −0.016 |
| | (−1.07) | (−3.26) | (−0.43) | (−0.55) | (0.92) | (−0.67) |
| $D_MED_{t+4}$ | −0.011 | −0.052** | 0.000 | 0.012 | 0.043* | −0.000 |
| | (−0.41) | (−2.19) | (0.01) | (0.57) | (1.76) | (−0.00) |
| $D_MED_{t+5}$ on | 0.009 | −0.083*** | −0.004 | −0.014 | 0.020 | −0.029 |
| | (0.38) | (−3.24) | (−0.10) | (−0.86) | (0.80) | (−0.95) |
| $D_REC_{t}$ | −0.007 | −0.045 | −0.046 | −0.026 | 0.025 | 0.035 |
| | (−0.11) | (−1.05) | (−1.13) | (−0.61) | (0.80) | (1.17) |
| $D_REC_{t+1}$ | −0.033 | −0.057 | −0.057 | 0.023 | 0.047 | 0.061 |
| | (−0.94) | (−1.46) | (−1.39) | (0.42) | (0.88) | (1.45) |
| $D_REC_{t+2}$ | −0.045 | −0.075 | −0.082 | 0.078 | 0.109* | 0.101** |
| | (−1.30) | (−1.24) | (−1.49) | (1.59) | (1.94) | (2.03) |
| $D_REC_{t+3}$ | −0.006 | −0.025 | −0.038 | 0.113*** | 0.180*** | 0.175*** |
| | (−0.13) | (−0.38) | (−0.66) | (4.30) | (4.11) | (3.14) |
| $D_REC_{t+4}$ | 0.073** | −0.064* | −0.013 | 0.089*** | 0.149*** | 0.197*** |
| | (2.01) | (−1.69) | (−0.19) | (3.71) | (2.80) | (3.14) |
| $D_REC_{t+5}$ on | 0.095** | 0.002 | 0.065 | 0.146*** | 0.211*** | 0.298*** |
| | (2.22) | (0.04) | (0.77) | (3.98) | (2.97) | (3.77) |
| adj. $R^2$ | 0.9177 | 0.9675 | 0.9754 | 0.9348 | 0.9583 | 0.9741 |
| F-test for D_MED | 0.57 | 3.29 | 0.37 | 0.37 | 0.78 | 0.51 |
| (*p*-value) | (0.72) | (0.01) | (0.87) | (0.87) | (0.56) | (0.77) |
| F-test for D_REC | 3.10 | 1.93 | 4.91 | 5.88 | 4.12 | 3.50 |
| (*p*-value) | (0.01) | (0.09) | (0.00) | (0.00) | (0.00) | (0.00) |
| F-test for all | 1.92 | 2.75 | 2.52 | 3.11 | 2.60 | 2.07 |
| (*p*-value) | (0.04) | (0.00) | (0.01) | (0.00) | (0.00) | (0.02) |
| | 1. Murder | | | 1. Burglary | | |
| $D_MED_{t-2}$ | 0.046 | −0.001 | −0.028 | 0.024 | 0.022 | 0.026* |
| | (1.60) | (−0.02) | (−0.83) | (1.28) | (1.38) | (1.78) |
| $D_MED_{t-1}$ | 0.053 | −0.023 | −0.057 | 0.020 | 0.007 | 0.012 |
| | (1.52) | (−0.62) | (−1.28) | (0.84) | (0.35) | (0.60) |
| $D_REC_{t-2}$ | −0.102** | −0.118*** | −0.097** | −0.013 | 0.024 | −0.017 |
| | (−2.06) | (−2.68) | (−2.12) | (−0.34) | (0.84) | (−0.65) |
| $D_REC_{t-1}$ | −0.050 | −0.112** | −0.083 | −0.043 | −0.001 | −0.009 |
| | (−1.06) | (−2.25) | (−1.60) | (−1.00) | (−0.04) | (−0.28) |
| F-test for D_MED | 2.06 | 0.21 | 0.85 | 1.06 | 0.95 | 1.61 |
| (*p*-value) | (0.13) | (0.81) | (0.43) | (0.35) | (0.39) | (0.20) |
| F-test for D_REC | 2.35 | 4.60 | 2.39 | 0.53 | 0.40 | 0.21 |
| (*p*-value) | (0.10) | (0.01) | (0.09) | (0.59) | (0.67) | (0.81) |
| F-test for all | 2.21 | 2.36 | 1.43 | 0.78 | 0.64 | 0.97 |
| (*p*-value) | (0.07) | (0.05) | (0.22) | (0.54) | (0.63) | (0.42) |
| $D_MED_{t}$ | 0.010 | −0.063 | −0.107** | 0.014 | 0.008 | 0.013 |
| | (0.27) | (−1.61) | (−2.21) | (0.56) | (0.39) | (0.68) |
| $D_MED_{t+1}$ | −0.009 | −0.087** | −0.137** | 0.018 | 0.005 | 0.012 |
| | (−0.23) | (−2.09) | (−2.57) | (0.60) | (0.24) | (0.52) |
| $D_MED_{t+2}$ | 0.011 | −0.076* | −0.134** | 0.009 | 0.002 | 0.015 |
| | (0.28) | (−1.89) | (−2.28) | (0.31) | (0.08) | (0.61) |
| $D_MED_{t+3}$ | 0.023 | −0.081* | −0.131** | −0.020 | −0.016 | 0.014 |
| | (0.55) | (−1.84) | (−2.00) | (−0.84) | (−0.70) | (0.50) |
| $D_MED_{t+4}$ | 0.035 | −0.067 | −0.131* | −0.005 | −0.003 | 0.027 |
| | (0.74) | (−1.36) | (−1.73) | (−0.19) | (−0.10) | (0.80) |
| $D_MED_{t+5}$ on | 0.000 | −0.169*** | −0.229** | −0.028 | −0.051 | −0.001 |
| | (0.01) | (−3.07) | (−2.49) | (−1.35) | (−1.58) | (−0.01) |
| $D_REC_{t}$ | −0.009 | −0.096* | −0.036 | −0.039 | −0.008 | 0.007 |
| | (−0.17) | (−1.80) | (−0.56) | (−0.65) | (−0.19) | (0.19) |
| $D_REC_{t+1}$ | −0.099* | −0.232*** | −0.146* | 0.058 | 0.064 | 0.067 |
| | (−1.82) | (−3.92) | (−1.88) | (0.66) | (0.94) | (1.28) |
| $D_REC_{t+2}$ | −0.169** | −0.339*** | −0.240** | 0.134** | 0.124* | 0.098* |
| | (−2.20) | (−4.00) | (−2.18) | (2.00) | (1.86) | (1.67) |
| $D_REC_{t+3}$ | −0.021 | −0.244*** | −0.207* | 0.138*** | 0.180*** | 0.174*** |
| | (−0.40) | (−2.76) | (−1.73) | (2.64) | (2.76) | (2.60) |
| $D_REC_{t+4}$ | 0.053 | −0.194** | −0.147 | 0.125*** | 0.127* | 0.193** |
| | (1.07) | (−2.24) | (−1.07) | (2.63) | (1.91) | (2.42) |
| $D_REC_{t+5}$ on | −0.002 | −0.231** | −0.187 | 0.190*** | 0.208** | 0.310*** |
| | (−0.04) | (−2.07) | (−1.03) | (5.51) | (2.58) | (3.22) |
| adj. $R^2$ | 0.9017 | 0.9182 | 0.9216 | 0.9397 | 0.9625 | 0.9771 |
| F-test for D_MED | 0.21 | 2.07 | 1.64 | 0.73 | 1.09 | 0.43 |
| (*p*-value) | (0.96) | (0.07) | (0.15) | (0.60) | (0.36) | (0.83) |

(*continued on next page*)

**Table 4** (*continued*).

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| F-test for D_REC | 2.30 | 4.26 | 1.84 | 6.46 | 1.92 | 2.64 |
| (*p*-value) | (0.04) | (0.00) | (0.10) | (0.00) | (0.09) | (0.02) |
| F-test for all | 1.25 | 3.04 | 1.75 | 3.66 | 2.10 | 1.81 |
| (*p*-value) | (0.26) | (0.00) | (0.07) | (0.00) | (0.02) | (0.06) |
| | 2. Rape | | | 2. Larceny Theft | | |
| $D_MED_{t-2}$ | 0.003 | −0.002 | −0.024 | 0.015 | 0.018 | 0.006 |
| | (0.10) | (−0.10) | (−1.18) | (1.09) | (1.37) | (0.51) |
| $D_MED_{t-1}$ | 0.031 | 0.028 | −0.003 | 0.003 | 0.004 | −0.016 |
| | (1.08) | (1.28) | (−0.11) | (0.22) | (0.24) | (−1.11) |
| $D_REC_{t-2}$ | 0.037 | 0.026 | −0.018 | −0.014 | 0.025 | −0.025 |
| | (1.10) | (0.75) | (−0.63) | (−0.61) | (1.54) | (−1.35) |
| $D_REC_{t-1}$ | 0.055 | 0.068 | 0.025 | −0.039 | 0.020 | −0.009 |
| | (1.16) | (1.26) | (0.61) | (−1.44) | (0.85) | (−0.37) |
| F-test for D_MED | 0.58 | 0.94 | 0.84 | 0.60 | 0.94 | 1.21 |
| (*p*-value) | (0.56) | (0.39) | (0.43) | (0.55) | (0.39) | (0.30) |
| F-test for D_REC | 1.15 | 0.95 | 0.65 | 1.14 | 1.25 | 0.96 |
| (*p*-value) | (0.32) | (0.39) | (0.52) | (0.32) | (0.29) | (0.38) |
| F-test for all | 0.84 | 0.90 | 0.72 | 0.89 | 0.98 | 1.05 |
| (*p*-value) | (0.50) | (0.47) | (0.58) | (0.47) | (0.42) | (0.38) |
| $D_MED_t$ | 0.071** | 0.073*** | 0.032 | 0.002 | 0.008 | −0.019 |
| | (2.38) | (2.85) | (1.05) | (0.14) | (0.56) | (−1.18) |
| $D_MED_{t+1}$ | 0.044 | 0.057** | 0.022 | 0.008 | 0.018 | −0.022 |
| | (1.51) | (2.17) | (0.75) | (0.46) | (1.14) | (−1.27) |
| $D_MED_{t+2}$ | 0.033 | 0.055** | 0.017 | 0.007 | 0.022 | −0.025 |
| | (1.20) | (2.31) | (0.52) | (0.41) | (1.41) | (−1.35) |
| $D_MED_{t+3}$ | 0.006 | 0.065** | 0.024 | 0.003 | 0.017 | −0.026 |
| | (0.20) | (2.28) | (0.67) | (0.16) | (0.86) | (−1.18) |
| $D_MED_{t+4}$ | 0.007 | 0.068** | 0.015 | 0.024 | 0.037* | −0.012 |
| | (0.20) | (2.01) | (0.36) | (1.21) | (1.71) | (−0.47) |
| $D_MED_{t+5}$ on | −0.013 | 0.077** | 0.017 | −0.004 | 0.029 | −0.039 |
| | (−0.49) | (2.48) | (0.36) | (−0.24) | (1.21) | (−1.38) |
| $D_REC_t$ | 0.031 | 0.045 | −0.010 | −0.009 | 0.053* | 0.026 |
| | (0.59) | (0.73) | (−0.24) | (−0.24) | (1.70) | (0.90) |
| $D_REC_{t+1}$ | 0.082 | 0.080 | 0.084 | 0.024 | 0.062 | 0.038 |
| | (1.36) | (1.46) | (1.59) | (0.49) | (1.20) | (1.00) |
| $D_REC_{t+2}$ | 0.022 | 0.039 | 0.007 | 0.083* | 0.138** | 0.078* |
| | (0.36) | (0.67) | (0.12) | (1.80) | (2.31) | (1.67) |
| $D_REC_{t+3}$ | −0.076 | 0.035 | 0.033 | 0.084*** | 0.191*** | 0.134*** |
| | (−0.87) | (0.64) | (0.41) | (3.35) | (4.87) | (2.63) |
| $D_REC_{t+4}$ | −0.007 | 0.087 | 0.106 | 0.059*** | 0.173*** | 0.136** |
| | (−0.06) | (1.56) | (1.23) | (2.81) | (3.96) | (2.38) |
| $D_REC_{t+5}$ on | −0.023 | 0.124** | 0.182* | 0.127*** | 0.242*** | 0.212*** |
| | (−0.27) | (2.05) | (1.66) | (4.19) | (3.93) | (2.92) |
| adj. $R^2$ | 0.8426 | 0.9020 | 0.9286 | 0.9248 | 0.9534 | 0.9709 |
| F-test for D_MED | 1.13 | 1.64 | 0.19 | 0.48 | 0.69 | 0.70 |
| (*p*-value) | (0.34) | (0.15) | (0.97) | (0.79) | (0.63) | (0.62) |
| F-test for D_REC | 0.62 | 1.60 | 3.29 | 4.47 | 6.55 | 2.63 |
| (*p*-value) | (0.69) | (0.16) | (0.01) | (0.00) | (0.00) | (0.02) |
| F-test for all | 0.86 | 1.23 | 1.80 | 2.34 | 3.64 | 1.69 |
| (*p*-value) | (0.57) | (0.27) | (0.06) | (0.01) | (0.00) | (0.08) |
| | 3. Robbery | | | 3. Auto Vehicle Theft | | |
| $D_MED_{t-2}$ | 0.012 | −0.020 | −0.039** | −0.050 | 0.026 | 0.001 |
| | (0.47) | (−1.04) | (−1.99) | (−1.44) | (1.14) | (0.04) |
| $D_MED_{t-1}$ | 0.042 | −0.004 | −0.034 | −0.056 | 0.021 | −0.013 |
| | (1.11) | (−0.14) | (−1.17) | (−1.60) | (0.86) | (−0.49) |
| $D_REC_{t-2}$ | −0.006 | −0.002 | −0.054 | −0.124 | −0.089** | −0.059 |
| | (−0.10) | (−0.05) | (−1.63) | (−1.64) | (−2.15) | (−1.30) |
| $D_REC_{t-1}$ | −0.042 | −0.063 | −0.092** | −0.114 | −0.116*** | 0.002 |
| | (−0.65) | (−1.35) | (−2.20) | (−1.61) | (−3.00) | (0.04) |
| F-test for D_MED | 0.67 | 0.56 | 1.98 | 1.96 | 0.77 | 0.15 |
| (*p*-value) | (0.51) | (0.57) | (0.14) | (0.14) | (0.46) | (0.86) |
| F-test for D_REC | 0.21 | 0.94 | 2.76 | 2.42 | 5.42 | 1.09 |
| (*p*-value) | (0.81) | (0.39) | (0.06) | (0.09) | (0.00) | (0.34) |
| F-test for all | 0.46 | 0.74 | 2.40 | 2.10 | 3.38 | 0.61 |
| (*p*-value) | (0.77) | (0.56) | (0.05) | (0.08) | (0.01) | (0.66) |
| $D_MED_t$ | 0.022 | −0.026 | −0.064** | −0.110*** | −0.020 | −0.056* |
| | (0.63) | (−1.00) | (−2.18) | (−3.20) | (−0.78) | (−1.90) |
| $D_MED_{t+1}$ | 0.019 | −0.039 | −0.089** | −0.100** | 0.012 | −0.042 |
| | (0.54) | (−1.46) | (−2.53) | (−2.32) | (0.43) | (−1.24) |
| $D_MED_{t+2}$ | 0.002 | −0.049* | −0.107*** | −0.132*** | −0.004 | −0.064 |
| | (0.05) | (−1.66) | (−2.88) | (−2.82) | (−0.11) | (−1.58) |
| $D_MED_{t+3}$ | −0.033 | −0.066* | −0.120*** | −0.155*** | 0.023 | −0.023 |
| | (−1.08) | (−1.91) | (−2.64) | (−3.23) | (0.59) | (−0.53) |
| $D_MED_{t+4}$ | 0.008 | −0.032 | −0.106** | −0.139*** | 0.043 | −0.017 |
| | (0.28) | (−0.88) | (−2.15) | (−2.70) | (0.93) | (−0.33) |
| $D_MED_{t+5}$ on | 0.032 | −0.074* | −0.141** | −0.101** | −0.009 | −0.049 |
| | (0.98) | (−1.94) | (−2.49) | (−2.53) | (−0.21) | (−0.84) |
| $D_REC_t$ | −0.037 | −0.102** | −0.110*** | −0.074 | −0.103** | 0.070 |
| | (−0.65) | (−2.45) | (−2.59) | (−1.06) | (−2.06) | (1.23) |
| $D_REC_{t+1}$ | −0.026 | −0.105 | −0.119** | 0.000 | −0.109 | 0.083 |
| | (−0.49) | (−1.46) | (−2.22) | (0.00) | (−1.49) | (1.11) |
| $D_REC_{t+2}$ | 0.027 | −0.044 | −0.110 | 0.014 | −0.124 | 0.101 |
| | (0.51) | (−0.50) | (−1.48) | (0.13) | (−1.56) | (1.17) |
| $D_REC_{t+3}$ | 0.082** | 0.011 | −0.088 | 0.222*** | −0.009 | 0.222** |
| | (2.16) | (0.12) | (−1.12) | (3.68) | (−0.12) | (2.09) |
| $D_REC_{t+4}$ | 0.147** | −0.050 | −0.089 | 0.178 | −0.129 | 0.297** |
| | (2.17) | (−0.64) | (−0.96) | (1.40) | (−1.03) | (2.40) |

(*continued on next page*)

**Table 4** (*continued*).

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| D_REC$_{t+5}$on | 0.228*** | 0.070 | 0.053 | 0.080 | −0.221* | 0.397*** |
| | (5.16) | (0.91) | (0.49) | (0.62) | (−1.76) | (2.77) |
| adj. $R^2$ | 0.9641 | 0.9787 | 0.9850 | 0.9111 | 0.9594 | 0.9719 |
| F-test for D_MED | 0.85 | 1.08 | 1.84 | 3.90 | 0.43 | 0.83 |
| (*p*-value) | (0.51) | (0.37) | (0.10) | (0.00) | (0.83) | (0.53) |
| F-test for D_REC | 6.62 | 1.34 | 5.88 | 3.10 | 1.39 | 1.91 |
| (*p*-value) | (0.00) | (0.25) | (0.00) | (0.01) | (0.23) | (0.09) |
| F-test for all | 4.46 | 1.51 | 3.69 | 3.97 | 0.94 | 1.30 |
| (*p*-value) | (0.00) | (0.13) | (0.00) | (0.00) | (0.50) | (0.22) |
| | 4. Aggravated Assault | | | | | |
| D_MED$_{t-2}$ | 0.026 | −0.039* | −0.006 | | | |
| | (0.71) | (−1.77) | (−0.26) | | | |
| D_MED$_{t-1}$ | 0.022 | −0.056*** | −0.004 | | | |
| | (0.67) | (−2.82) | (−0.19) | | | |
| D_REC$_{t-2}$ | 0.029 | 0.031 | 0.045 | | | |
| | (0.52) | (1.12) | (1.44) | | | |
| D_REC$_{t-1}$ | −0.024 | −0.007 | −0.013 | | | |
| | (−0.42) | (−0.22) | (−0.43) | | | |
| F-test for D_MED | 0.42 | 4.43 | 0.04 | | | |
| (*p*-value) | (0.66) | (0.01) | (0.96) | | | |
| F-test for D_REC | 0.24 | 0.89 | 2.43 | | | |
| (*p*-value) | (0.79) | (0.41) | (0.09) | | | |
| F-test for all | 0.33 | 2.71 | 1.26 | | | |
| (*p*-value) | (0.86) | (0.03) | (0.28) | | | |
| D_MED$_t$ | 0.008 | −0.071*** | −0.002 | | | |
| | (0.25) | (−3.56) | (−0.07) | | | |
| D_MED$_{t+1}$ | 0.016 | −0.074*** | 0.012 | | | |
| | (0.45) | (−3.36) | (0.42) | | | |
| D_MED$_{t+2}$ | 0.007 | −0.090*** | 0.016 | | | |
| | (0.22) | (−3.60) | (0.48) | | | |
| D_MED$_{t+3}$ | −0.018 | −0.077*** | 0.030 | | | |
| | (−0.56) | (−3.05) | (0.87) | | | |
| D_MED$_{t+4}$ | −0.002 | −0.069** | 0.056 | | | |
| | (−0.06) | (−2.49) | (1.41) | | | |
| D_MED$_{t+5}$on | 0.013 | −0.100*** | 0.072 | | | |
| | (0.42) | (−2.96) | (1.46) | | | |
| D_REC$_t$ | −0.032 | −0.067 | −0.055 | | | |
| | (−0.45) | (−1.43) | (−1.06) | | | |
| D_REC$_{t+1}$ | −0.066 | −0.075 | −0.089* | | | |
| | (−1.26) | (−1.61) | (−1.72) | | | |
| D_REC$_{t+2}$ | −0.062 | −0.098 | −0.105 | | | |
| | (−1.29) | (−1.35) | (−1.64) | | | |
| D_REC$_{t+3}$ | −0.027 | −0.044 | −0.049 | | | |
| | (−0.51) | (−0.51) | (−0.66) | | | |
| D_REC$_{t+4}$ | 0.047 | −0.105* | −0.028 | | | |
| | (1.01) | (−1.88) | (−0.33) | | | |
| D_REC$_{t+5}$on | 0.061 | −0.046 | 0.041 | | | |
| | (1.46) | (−0.80) | (0.38) | | | |
| adj. $R^2$ | 0.8952 | 0.9553 | 0.9691 | | | |
| F-test for D_MED | 0.27 | 3.18 | 1.28 | | | |
| (*p*-value) | (0.93) | (0.01) | (0.27) | | | |
| F-test for D_REC | 1.95 | 1.26 | 3.29 | | | |
| (*p*-value) | (0.08) | (0.28) | (0.01) | | | |
| F-test for all | 1.14 | 2.32 | 2.13 | | | |
| (*p*-value) | (0.33) | (0.01) | (0.02) | | | |

The Table summarizes the tests for parallel trends one and two periods before marijuana legalization. Models (1) to (3) are for Violent Crimes. Models (4) to (6) are for Property Crimes. All models include state and year fixed-effects, as well as state-level covariates. t-values are in parentheses for the main estimation, while p-values are in parentheses for F-tests of joint significance. Models (2) and (5) add state specific time trends by multiplying the State dummy by $t$. Models (3) and (6) add a state-specific trend $t^2$ in addition to $t$. The sample period is from 1995 to 2019 for 50 states. "t-2" and "t-1" correspond to 2 years and 1 year before marijuana legalization.

*** p-values smaller than 0.01.

** p-values smaller than 0.05.

* p-values smaller than 0.1.

### 4.3. Synthetic difference-in-differences

Thus far we have used standard estimation approaches based on difference in differences, in order to tease out salient impacts of marijuana legalization. A key assumption underlying such approaches is that of parallel trends: the average outcome for the treated group should be roughly parallel to the average for the control group. This enables the researcher to argue that post-treatment differences between the groups causally reflect the treatment. If the parallel trend assumption fails, it can lead to biased and unreliable estimates (Hansen, 2022, pp. 677–679). Some common strategies to ameliorate such challenges involve using robust standard errors and adjusting for covariates – the latter helping to control for time-varying confounders – both of which we utilized in our foregoing approaches. A more sophisticated method to handle potential violations of the parallel trends assumption involves constructing synthetic control groups, that is, weighted combinations of untreated units that resemble the treatment group. After checking for parallel trends, we therefore adopt a synthetic approach.

Table 4 presents our results on parallel trends. In particular, we add dummies for one and two years before legalization— $t-1$ and $t-2$, respectively. If the estimated coefficients on $t-1$ and $t-2$ are insignificant, this provides evidence in favor of parallel trends. As before, we focus on the most comprehensive models, in columns (3) and (6). Let us first consider violent crimes overall, focusing on model (3) and the estimated coefficients for the first four rows of dummy variables– $D_MED_{t-2}$, $D_MED_{t-1}$, $D_REC_{t-2}$, and $D_REC_{t-1}$. For both medical and recreational legalization the estimated coefficients are all insignificant. When we look at property crimes overall in model (6), we observe the same pattern of insignificant coefficients. This indicates that for crimes as a whole there is evidence of parallel trends. We dig deeper in Table 4 and examine the coefficients for individual crimes. Since there are a total of four dummy variables for each crime and four violent crime subcategories, the total number of estimates

**Table 5**
Synthetic difference-in-differences Estimation.

| | Medical Legalization | | Recreational Legalization | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| 1. Violent Crimes | | | | |
| | −0.0050 | −0.0965 | 0.0384 | −0.1741** |
| | (−0.10) | (−1.42) | (0.97) | (−2.18) |
| 2. Property Crimes | | | | |
| | −0.0366* | −0.2093* | 0.0753** | −0.0918 |
| | (−1.74) | (−1.71) | (2.30) | (−0.47) |
| 3. Murder | | | | |
| | −0.1479*** | −0.0481 | 0.0064 | −0.0632 |
| | (−2.71) | (−0.42) | (0.10) | (−0.94) |
| 4. Rape | | | | |
| | 0.0275 | −0.0466 | 0.0281 | 0.1038 |
| | (0.53) | (−0.62) | (0.49) | (0.77) |
| 5. Robbery | | | | |
| | −0.0525 | 0.0118 | 0.0744 | 0.2028 |
| | (−0.86) | (0.06) | (1.17) | (0.55) |
| 6. Aggravated Assault | | | | |
| | 0.0042 | −0.1357* | 0.0069 | −0.3699** |
| | (0.08) | (−1.87) | (0.18) | (−2.38) |
| 7. Burglary | | | | |
| | −0.0613 | −0.2743* | 0.0973 | −0.1289 |
| | (−1.45) | (−1.88) | (1.57) | (−0.46) |
| 8. Larceny Theft | | | | |
| | −0.0164 | −0.1764 | 0.0715*** | −0.0634 |
| | (−0.68) | (−1.54) | (2.82) | (−0.32) |
| 9. Auto Vehicle Theft | | | | |
| | −0.0844 | −0.2983*** | 0.1169 | −0.1930 |
| | (−1.47) | (−2.98) | (1.36) | (−1.04) |
| State and Year Fixed Effect | Yes | Yes | Yes | Yes |
| State Specific Time Trend | | Yes | | Yes |

The Table presents results for the impact of marijuana legalization on violent and property crimes. We analyze two types of legalization, for Medical marijuana and Recreational marijuana usage. The estimation is the synthetic difference-in-differences approach developed by Arkhangelsky et al. (2021). This approach combines synthetic controls and difference-in-differences methods. The main statistic is the estimated effect of the treatment. States that legalized marijuana are considered as treatment groups. The variance of the effect is estimated with the bootstrap procedure. t-values are in parentheses.

*** p-values smaller than 0.01.

** p-values smaller than 0.05.

* p-values smaller than 0.1.

for violent crimes is sixteen. Similarly for property crime with three subcategories, the total number of estimates for property crimes is twelve. Here the pattern is still strong but more nuanced: for violent crimes thirteen coefficients are insignificant and three are significant; for property crimes eleven coefficients are insignificant and only one is significant. Thus, at the individual crime level there is evidence that parallel trends hold in many but not all cases, while at the overall crime level there is evidence in favor of parallel trends. Given this evidence, we proceed with a synthetic difference-in-differences model.

We use the framework of synthetic difference-in-differences, which combines synthetic control and difference-in-differences (Arkhangelsky et al., 2021). Similar to traditional difference in differences, the synthetic approach permits treated and control units to trend on different levels prior to a reform. And similar to synthetic control, it optimally constructs an artificial matched control unit, thereby reducing the need for assuming parallel trends. This method weakens but does not eliminate the parallel trend type of requirement.

While the synthetic difference in differences approach obviates the need for the standard parallel trends assumption, it does have requirements, the more important of which follow.[5] First, from a practical perspective it requires a balanced panel of data. Second, no units should have been exposed to treatment at the start of the sample, and some units must never have been treated. Third, for bootstrap inference the number of treated units should exceed one. Fourth, from a theoretical perspective, the variance and confidence interval results are asymptotic and therefore only hold approximately; thus the implementation requires a large enough sample size. Arkhangelsky et al. (2021) perform simulation diagnostics which indicate that the estimator performs well with as few as one treatment unit and with a panel of 50 states and 40 time periods. Our sample is roughly comparable to this benchmark. Fifth, a crucial aspect of estimation is to balance having enough periods to generate synthetic cohorts, with avoiding nonstationarities in the treatment outcomes. This latter requirement is particularly important for us, since there has been a significant downward shift in most crimes over our sample—with the exception of rape, which had a significant increase. Such patterns suggest a potential violation of the synthetic difference in differences condition just discussed. In order to deal with this potential challenge, we incorporate not only fixed effects but state specific time trends as a precaution.

We report our synthetic difference-in-differences results in Table 5. As before, we focus on the most comprehensive models—columns (2) and (4)—which provide for both fixed effects and a state specific time trend. All models include covariates. As expected, the addition of state specific time trends sometimes modifies the results relative to

[5] For further discussion of requirements for synthetic approaches, see Abadie (2021) and Athey et al. (2024).

models without such trends. This is evidence to support their inclusion in the models. For medical marijuana effects in column (2) we find an insignificant effect for violent crimes overall. Nevertheless, the violent crime subcategory of aggravated assault has a significant negative effect, again underscoring the importance of breaking down the crime rates into individual components. We also see a negative and significant effect for property crime, driven by burglary and vehicle theft.

Our benchmark results for recreational marijuana legalization are in column (4). While the results for property crimes are insignificant overall and within subcategories, those for violent crimes are significant overall, and again driven by the effect on aggravated assault. Thus, our synthetic difference-in-differences approach reveals that property crimes are an important conduit for the effect of medical marijuana legalization, while violent crimes are reflected in recreational legalization. In addition, the significance of aggravated assault, the most frequent violent crime, is apparent.

### 4.4. Discussion

We have pursued our leading question on the consequences of marijuana legalization with a variety of empirical techniques, culminating in a synthetic difference in differences framework. Such variety is necessary to tease out ramifications that might not be clear using a single approach, and to respect potential data limitations.

Our first approach uses difference in differences. Both recreational and medical marijuana legalization have negative effects, the coefficients go in the expected direction of the (Becker, 1968) framework, where legalization make other criminal activities unprofitable. Though the overall effect of medical legalization is insignificant, the effect on both murder and robbery is negative. In the case of rape, the positive effect is initially puzzling. However, the expanded definition of rape in the data starting in 2012 could go a long way to explain it. In addition, there has been an increase in reporting rape during the last four decades, due to changing social norms and the #MeToo movement (Baumer et al., 2003; Stewart et al., 2024; Wolitzky-Taylor et al., 2011). In the case of recreational legalization, the negative effect on crime is dominated by aggravated assault. This is likely a reflection of the psychopharmacological channel.

As far as property crime, medical marijuana has a small negative effect on motor vehicle theft; but for recreational marijuana, property crime, and larceny and vehicle theft increase after its legalization. Like previous studies (Pacula and Smart, 2017; Lee, 2025) our analysis does not narrow down the precise mechanism. As in previous research (Lee, 2025) there are three plausible channels for the increase in crime following legalization: addicted individuals who turn to illicit activity to fund marijuana abuse (economic channel); loyalty to existing illicit suppliers (systemic channel); and cognitive impairment which leads to criminal activity (psychopharmacological channel).

Our second approach considers a dynamic model that accounts for implementation lags, and yields broadly similar results. Here we find that lag effects matter for both violent and property crimes. For overall crime rates, the effects come from recreational legalization. For specific violent crime categories, medical legalization in a given year has an effect, namely, in reducing robbery and murder in subsequent years. This salutary result is consistent with the Becker mechanism, and also reflects the fact that establishing a formal marijuana market—to replace the crime-ridden black market—takes time. For property crimes, both overall and in specific categories, the lag effects come from recreational legalization. Such crimes increase over time, but we argue that this is also consistent with the economic, systemic, or psychopharmacological channels.

In our main analysis, we use synthetic difference in differences estimation. When results are significant, the coefficients are always negative. For medical legalization, there is a significant effect on property crimes overall, and on burglary and vehicle theft. This could reflect the Becker hypothesis, or perhaps, as others have discussed, reduced aggression from marijuana use (Boles and Miotto, 2003). Similarly, we find that legalization of recreational marijuana reduces aggravated assault. As above, we cannot assign exact causality but previous literature discusses, for example, the existence of dispensaries creating a natural source of policing (Kepple and Freisthler, 2012).

What emerges from our multi-step analysis is a birds-eye view of legalization: medical and recreational legalization have different impacts and operate through diverse channels, with significant lag effects. This heterogeneity between the type of legalization supports the predictions of previous research, such as (Pacula and Smart, 2017) and Lee (2025).

Moving forward, we would encourage policymakers wishing to develop credible policy analysis and predictions for the impact of marijuana legalization to focus on the specific type of legalization (recreational or medical), and to draw lessons from states with similar characteristics to their own state.

## 5. Conclusions

We examine the relationship between marijuana legalization and crime rates, using state level panel data that we compile from a large variety of sources. Since legalization occurred in a staggered fashion across states, we can study interesting dynamics in the impact on crime. While previous research tends to focus on a few states, we consider all 50 US states. And while previous research typically examines either recreational marijuana (Dragone et al., 2019; Sabia et al., 2024; Lee, 2025) or medical marijuana (Pacula and Chaloupka, 2002; Pacula and Smart, 2017), we integrate both recreational and medical legalization in one empirical framework. This is a major contribution of our research. A second contribution is that when analyzing all US states, we show that the heterogeneity effect of Pacula and Smart (2017) for medical marijuana also applies to recreational marijuana.

The overarching result from our main synthetic difference in differences model is that medical legalization reduces property crime, while recreational legalization reduces violent crime. Such effects support the Becker hypothesis that legalization drives out crime. Building up to the synthetic difference in difference model, we discovered that there may also be important lag effects. The diverse and potentially time-varying impacts of medical and recreational legalization raise a cautionary note to policymakers: those considering legalization should wait a few years before pronouncing on the cost–benefit impact, focus on the specific type of legalization, and study closely the outcomes from similar states.

The diversity of our results for both medical and recreational marijuana echoes related research which studied only one type of marijuana legalization: (Pacula and Smart, 2017) underscore the challenge in attributing a single mechanism to the impact of legalization of medical marijuana, while (Dragone et al., 2019) and Lee (2025) voice similar challenges for recreational marijuana. Parsing the implications of such results will provide a rich soil for future research. Our finding that the medical and recreational effects go in different directions for the same crimes suggests that a particularly fruitful research avenue involves clarifying the net impact of legalizing medical and recreational marijuana for specific types of crime.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix

See Table A.1

**Table A.1**
Data collection by source.

| Variable | Source | Other |
|---|---|---|
| 1. Crime Data | FBI Uniform Crime Reports | Collected from Table 5 of the report each year, from 1995 to 2019 for all 50 states and DC. Variables include population; violent crime; property crime; murder; forcible rape; robbery; aggravated assault; burglary; larceny-theft; and motor vehicle theft. Violent crimes are the sum of murder, forcible rape, robbery, and aggravated assault. Property crimes offenses are burglary, larceny-theft, and motor vehicle theft. |
| 2. Governor's Party Affiliation | Ballotpedia.org and hand-collecting | When there is more than one governor from different parties in the same calendar year, the governor whose tenure exceeds six months is chosen. |
| 3. Majority Party of Senate & House | Ballotpedia.org | Nebraska has only Senate since 1937. |
| 4. Income per capita | Bureau of Economic Analysis | GDP per capita started in 1997, so we use personal income per capita (SAINC1). |
| 5. Unemployment Rate | Bureau of Labor Statistics | |
| 6. Police Force | FBI Uniform Crime Reports | Table 77. Variables include population, and officers. We calculate the number of police per 100,000 residents. |
| 7. Arrest Data | FBI Uniform Crime Reports | Table 69. Some states do not submit data during the year. Drug, DUI, and liquor arrests are also available. |
| 8. Population Density | Census Bureau | We extract population, total area of state, and total land area of state. |
| 9. Poverty Rate | Census Bureau | Table 21, Historical Poverty Tables: People and Families, 1959–2020. Number of Poor and Poverty Rate. |
| 10. Youth Population | Census Bureau American Community Survey (ACS) | State level data are available from 2000 to 2020. |
| 11. Education Attainment | Census Bureau American Community Survey (ACS) | Proportion of people 18-24, and 25 and above with a high school degree. Data available 2010–2019. |
| 12. Cigarette Data | CDC | 5 variables retrieved: State tax per pack; cigarette sales per capita; Gross cigarette tax revenue; price per pack; Federal tax per pack. Available 1970–2019. |
| 13. Marijuana Legalization | Handcollected from Wiki & NORML.org | Collected data for the following events, for recreational and medical marijuana: date the law was passed; date the law became effective; date the first legal dispensary opened. We also collected the effective date that CBD oil was made legal; and effective date of decriminalization. |
| 14. Neighbor State Effects | Handcollected from Wiki | We construct 2 variables, Avg_Med and Avg_Rec, which represent the fraction of neighbor states that have legalized medical and recreational marijuana. |

## Data availability

Mendeley

How Does Marijuana Legislation Affect Crime? Medical and Recreatio nal Laws Across 50 States (Reference data) (Mendeley Data)

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